PAPER_242 — Rings of Relativity: Einstein Ring Lensing Amplification in the Full MUGE

Date: October 2025 ## GAL-CLUS-022058s — Static Einstein-Ring Lensing Factor in the Master Universal Gravity Equation

Author: Daniel T. Murphy
Framework: Unified Quantum Field Framework (UQFF) v4.10
Calculator Class: `RingsOfRelativityEinsteinLensingMUGECalculator` (CP3, Session 60)
Encoded By: Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026
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Abstract

This paper presents the full Master Universal Gravity Equation (MUGE) for the GAL-CLUS-022058s gravitational lens system ("Rings of Relativity"). The key unique mathematical contribution is the **static Einstein ring lensing amplification factor** L_t, derived from the angular diameter distance ratio D_{LS}/D_S and the Schwarzschild radius of the galaxy cluster lens, applied as a multiplicative correction to the base gravitational field.

This is expressly **distinct** from the dynamic lensing modulation $L(t) = L_0 e^{-t/\tau} \cos(\omega t)$ already captured in CP3 class 81 (`UQFFLensingModulationRingsCalculator`, Session 53). The present paper captures the static, geometry-driven Einstein ring correction.

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2. System Parameters

Parameter	Symbol	Value	Units
Lens mass	M	$10^{14}\, M_\odot$	kg
Einstein radius	r	3.086×10^{20}	m (~10 kpc)
Redshift of lens	z_{lens}	0.5	dimensionless
Angular distance ratio	$L_{\text{factor}} = D_{LS}/D_S$	0.67	dimensionless
Magnetic field	B	10^{-5}	T
Gas velocity	v_{gas}	10^5	m/s
Oscillatory amplitude	A_{osc}	10^{-12}	m/s^2

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3. The Novel Lensing Amplification Term

3.1 Einstein Ring Lensing Factor

The static gravitational lensing amplification of the measured gravitational field along the line of sight through the Einstein ring geometry:

$$L_t = \frac{GM}{c^2 r} \cdot L_{\text{factor}}, \quad L_{\text{factor}} = \frac{D_{LS}}{D_S} = 0.67$$

$$\mathrm{corr}_L = 1 + L_t$$

where: - GM/c^2r is the dimensionless Schwarzschild-radius-to-Einstein-radius ratio - D_{LS} = angular diameter distance from lens to source - D_S = angular diameter distance from observer to source - The factor D_{LS}/D_S encodes the lensing geometry in the single-lens single-source approximation

3.2 Friedmann Correction at $z = 0.5$

$$H(z = 0.5) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} = H_0\sqrt{0.3 \cdot (1.5)^3 + 0.7}$$

This mid-redshift Friedmann correction is distinct from $z = 0$ (local), $z = 3.5$ (HUDF), or $z = 0.01$ (nearby lenses) used in earlier classes.

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4. Full MUGE: All Nine Terms

$$g_{\text{Rings}}(r,t) = \underbrace{T_1}_{\text{base+H+B+L}} + \underbrace{T_2}_{\text{UQFF}} + \underbrace{T_3}_{\Lambda} + \underbrace{T_4}_{\text{EM}} + \underbrace{T_q}_{\text{QM}} + \underbrace{T_{\text{fl}}}_{\text{fluid}} + \underbrace{T_{\text{osc}}}_{\text{2-mode}} + \underbrace{T_{\text{DM}}}_{\text{DM+pert2}} + \underbrace{T_{\text{wind}}}_{\text{stellar wind}}$$

Term 1 — Base gravity with H(z), magnetic, Einstein lensing:

$$T_1 = \frac{GM}{r^2}(1 + H(z)\,t)(1 - B/B_{\text{crit}})(1 + L_t)$$

Term 2 — UQFF unified field with time-reversal:

$$T_2 = (U_{g1} + U_{g4})(1 + f_{TRZ})$$

Term 3 — Cosmological constant:

$$T_3 = \frac{\Lambda c^2}{3}$$

Term 4 — Electromagnetic with vacuum density ratio:

$$T_4 = \frac{q v_{\text{gas}} B}{m_p}\left(1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}\right) \cdot s_{\text{EM}}, \quad \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} = \frac{7.09 \times 10^{-36}}{7.09 \times 10^{-37}} = 10$$

Term 5 — Quantum uncertainty:

$$T_q = \frac{\hbar}{\sqrt{\Delta x\,\Delta p}}\,\psi\,\frac{2\pi}{t_H}$$

Term 6 — Fluid (buoyancy-acceleration):

$$T_{\text{fl}} = \frac{\rho_{\text{fluid}}\,V\,g_{\text{base}}}{M}$$

Term 7 — Two-mode oscillation (standing wave + traveling wave):

$$T_{\text{osc}} = 2A\cos(kx)\cos(\omega t) + \frac{2\pi}{t_{\text{Gyr}}}A\cos(kx - \omega t)$$

The first mode is a **standing wave decomposition** ($2\cos\cos$); the second mode is a **Gyr-scaled traveling wave**. Together they form a beating-mode pair unique to this system.

Term 8 — Dark matter with $\delta_2 = 3GM/r^3$ perturbation:

$$T_{\text{DM}} = \frac{(M + M_{\text{DM}})(\delta\rho/\rho + 3GM/r^3)}{M}$$

The second-order perturbation $3GM/r^3$ is a tidal-force density correction distinct from density-contrast $\delta\rho/\rho$.

Term 9 — Stellar wind feedback:

$$T_{\text{wind}} = \frac{\rho_{\text{wind}}\, v_{\text{wind}}^2}{\rho_{\text{fluid}}}$$

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5. Distinction from Prior Art

Feature	Class 81 (Session 53)	This class (Session 60)
Lensing factor	$L(t) = L_0\, e^{-t/\tau} \cos(\omega_{\text{lens}} t)$	$L_t = (GM/c^2 r) \cdot (D_{LS}/D_S)$
Physics	Dynamic lens transit alignment	Static Einstein ring geometry
Time dependence	Yes (decaying oscillation)	No (geometric constant)
Parameter	$L_0 = 0.15$, τ_{lens}, ω_{lens}	$L_{\text{factor}} = 0.67$ (distance ratio)
Oscillation	None	Two-mode standing + traveling wave
DM pert2	Not present	$3GM/r^3$ tidal correction
Wind term	Not present	$\rho_w v_w^2/\rho_{\text{fl}}$

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6. Numerical Verification

Using default parameters ($M = 1.989 \times 10^{44}$ kg, $r = 3.086 \times 10^{20}$ m, $z = 0.5$):

$$L_t = \frac{(6.6743 \times 10^{-11})(1.989 \times 10^{44})}{(2.998 \times 10^8)^2 (3.086 \times 10^{20})} \times 0.67 \approx 1.6 \times 10^{-3}$$

$$\text{corr}_L \approx 1.0016 \qquad (\sim 0.16\%\,\text{amplification})$$

$$H(z = 0.5) = H_0 \sqrt{0.3 \times 3.375 + 0.7} \approx 1.27\, H_0$$

The lensing correction is small but non-zero and physically motivated by the Einstein ring geometry of GAL-CLUS-022058s ($\theta_E \approx 10''$, confirmed by HST imaging).

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7. Integration Into UQFF Pipeline

- **CP3 class:** `RingsOfRelativityEinsteinLensingMUGECalculator` (112th calculator, Session 60)
- **Aggregator:** v2.4.0 — registered in `CP3_CALCULATORS`
- **Source:** Doc 8 C++ class `RingsOfRelativity` (Grok/xAI, October 2025)
- **Complementary classes:**
  - Class 81 `UQFFLensingModulationRingsCalculator` — dynamic lens transit
  - Class 111 `NGC3603FullMUGECavityPressureCalculator` — companion new class (PAPER_243)

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8. References

- Hubble Space Telescope imaging of GAL-CLUS-022058s (Infante et al. 2022)
- Einstein (1936): Lens-star problem, Science 84, 506
- Schneider, Ehlers & Falco (1992): Gravitational Lenses, Springer
- Murphy, D.T. (2025): UQFF Manuscript — Doc 8 (Rings of Relativity MUGE, October 2025)
- Grok (xAI): C++ class `RingsOfRelativity`, encoded October 2025

PAPER_242 | Session 60 | CP3 class 111 (RingsOfRelativityEinsteinLensingMUGECalculator) | UQFF v4.10

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.109 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 5, \quad n_{\text{channel}} = 9/26$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 5$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits its sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^6 M_BH/M_⊙ yr** (quasinormal mode ringdown):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.109	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 51$2	✓ Sub-threshold